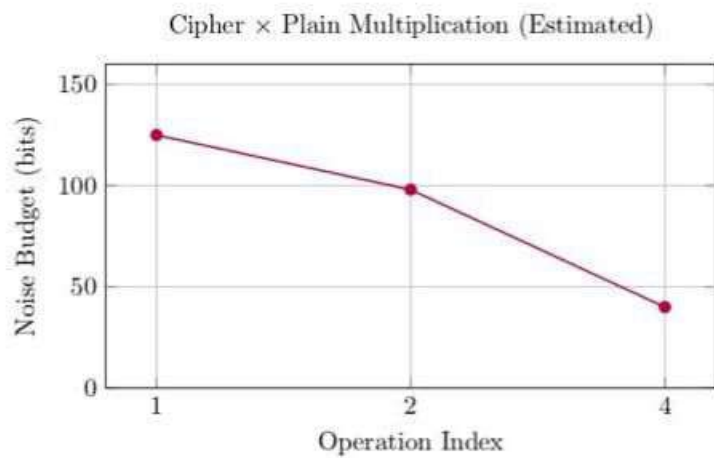
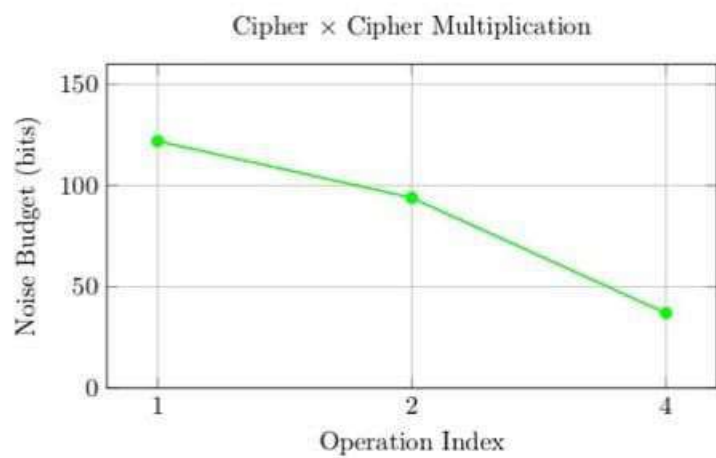
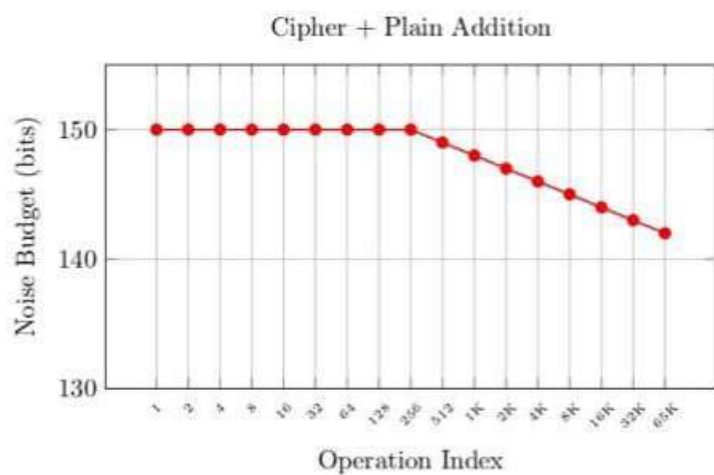
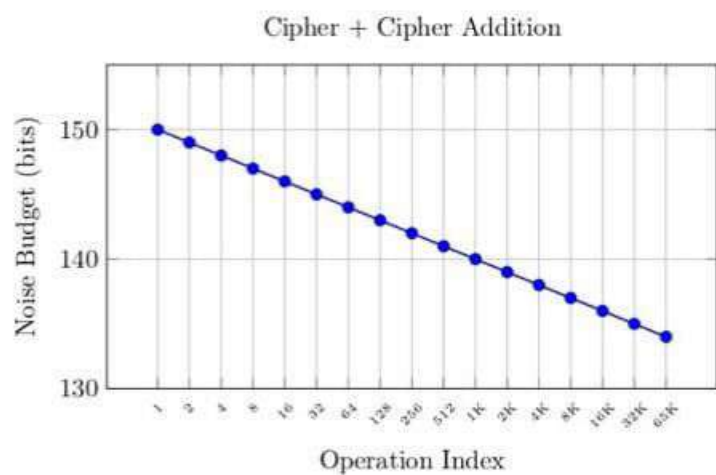




Noise budget analysis for different homomorphic operations. **Top-left:** Cipher+Cipher addition shows gradual noise increase. **Top-right:** Cipher+Plain addition maintains noise budget for longer. **Bottom-left:** Cipher $\times$ Cipher multiplication consumes noise budget rapidly. **Bottom-right:** Cipher $\times$ Plain multiplication (estimated) shows similar rapid noise consumption as Cipher $\times$ Cipher.